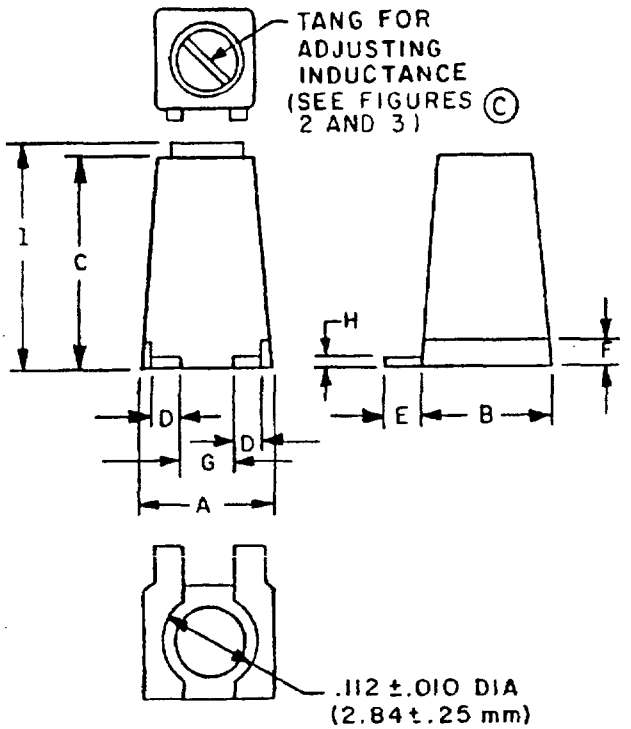
MILITARY SPECIFICATION SHEET

COILS, RADIO FREQUENCY, CHIP, VARIABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

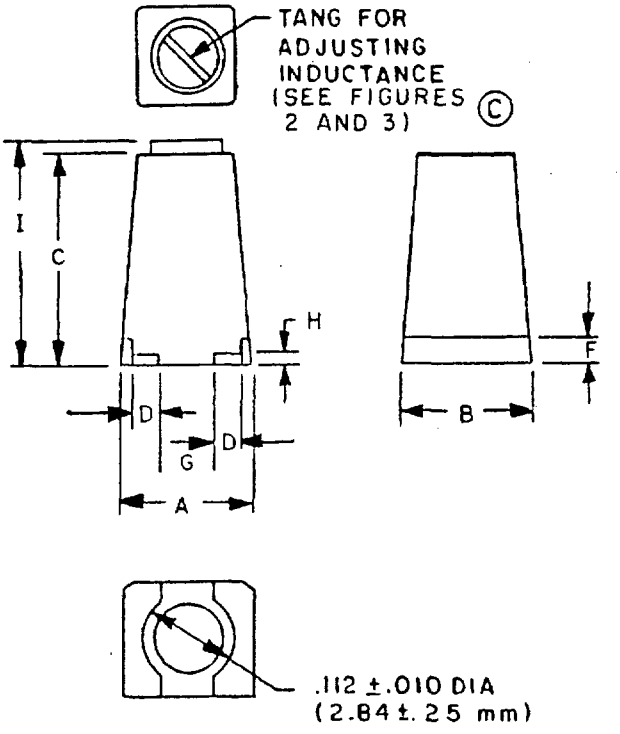
The requirements for acquiring the coils described herein shall consist of this specification and the latest issue of MIL-C-83446.



	INCHES	MM
A	.165 Max	4.19 Max
B	.130 Max	3.30 Max
C	.225 Max	5.72 Max
D	.030 ±.010	0.76 ±0.25
E	.040 +.015 -.010	1.02 +0.38 -0.25
F	.020 ±.005	0.51 ±0.13
G	.073 Min	1.85 Min
H	.005 ±.002	0.13 ±0.05
Ⓒ I	.250 Max	6.35 Max

For dash numbers 01 thru 47

(WITH TABS)



	INCHES	MM
A	.165 Max	4.19 Max
B	.130 Max	3.30 Max
C	.225 Max	5.72 Max
D	.030 ±.010	0.76 ±0.25
F	.020 ±.005	0.51 ±0.13
G	.073 Min	1.85 Min
H	.005 ±.002	0.13 ±0.05
Ⓒ I	.250 Max	6.35 Max

For dash numbers 48 thru 94

(WITHOUT TABS)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
Ⓒ 3. Dimension I is the maximum protrusion of the tang at minimum inductance.

FIGURE 1. Variable, chip coil.

Ⓒ denotes changes

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Weight: 0.5 gram maximum.

Operating temperature range: -55°C to +125°C.

Temperature rise (at 90°C): 35°C.

Maximum operating temperature: 125°C.

Altitude: 70,000 feet.

Dielectric withstanding voltage: Method 301 of MIL-STD-202, test voltage 700 volts rms.

Barometric pressure: Method 105, test condition C, MIL-STD-202 (70,000 feet), test voltage 200 volts rms.

Electrical characteristics (initial): See table I.

Electrical characteristics (final): See table II.

TABLE I. Electrical characteristics (initial) and dash numbers.

Dash number	Inductance			Q Min over L range	Q typical (at L nom)	Test frequency 1/	Self-resonant frequency (min)	DC resistance at 25°C (max)	Current (max) 2/	Incremental current (max) 3/
	Min	Nom	Max							
	µH	µH	µH			MHz	MHz	Ohms	mA	mA
01, 48	0.015	0.018	0.021	50	60	150	1200	0.03	750	---
02, 49	0.018	0.022	0.026	55	65	"	1100	0.03	"	---
03, 50	0.022	0.027	0.032	"	65	"	1000	0.03	"	---
04, 51	0.027	0.033	0.039	"	65	"	900	0.04	"	---
05, 52	0.033	0.039	0.047	"	65	"	750	0.04	"	---
06, 53	0.039	0.047	0.056	"	65	100	660	0.04	"	---
07, 54	0.047	0.056	0.068	"	65	"	610	0.05	700	---
08, 55	0.055	0.068	0.083	"	65	"	550	0.05	700	---
09, 56	0.067	0.082	0.099	"	65	"	520	0.06	645	---
10, 57	0.080	0.100	0.120	"	65	"	490	0.06	645	---
11, 58	0.096	0.120	0.144	50	60	50	460	0.07	600	---
12, 59	0.12	0.15	0.18	"	60	"	430	0.08	560	---
13, 60	0.14	0.18	0.22	"	60	"	385	0.09	525	---
14, 61	0.17	0.22	0.27	"	60	"	330	0.10	500	---
15, 62	0.21	0.27	0.33	"	60	"	300	0.15	400	---
16, 63	0.26	0.33	0.40	"	60	"	275	0.19	360	---
17, 64	0.32	0.39	0.47	40	50	"	240	0.25	300	---
18, 65	0.37	0.47	0.57	"	50	25	210	0.30	280	---
19, 66	0.46	0.56	0.68	"	50	"	190	0.40	250	---
20, 67	0.54	0.68	0.82	"	50	"	165	0.60	200	---
21, 68	0.68	0.82	1.00	"	50	"	155	0.70	180	---
22, 69	0.75	1.00	1.25	"	50	"	140	0.90	160	---
23, 70	0.90	1.20	1.50	"	50	"	125	1.1	150	---
24, 71	1.13	1.50	1.87	35	43	7.9	115	1.3	135	---
25, 72	1.35	1.80	2.25	35	43	"	105	1.7	120	---
26, 73	1.65	2.20	2.75	35	43	"	100	2.3	100	---
27, 74	2.02	2.70	3.37	25	35	"	85	3.0	90	---
28, 75	2.48	3.30	4.12	25	35	"	75	3.9	80	---

See footnotes at end of table.

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TABLE I. Electrical characteristics (initial) and dash numbers - Continued.

Dash number	Inductance			Q	Q	Test	Self-resonant	DC resistance	Current	Incremental current
	Min	Nom	Max	Min over L range	typical (at L nom)	frequency 1/	frequency (min)	at 25°C (max)	(max) 2/	(max) 3/
	µH	µH	µH			MHz	MHz	Ohms	mA	mA
29, 76	2.92	3.90	4.87	25	33	7.9	47	3.9	80	---
30, 77	3.52	4.70	5.88	25	33	"	30	3.9	80	---
31, 78	4.2	5.6	7.0	22	30	"	24	4.0	75	---
32, 79	5.1	6.8	8.5	"	30	"	23	4.0	75	---
33, 80	6.1	8.2	10.2	"	30	"	22	4.5	72	---
34, 81	7.5	10.0	12.5	"	30	"	21	4.5	72	---
35, 82	9.0	12.0	15.0	"	30	"	17	5.0	70	---
36, 83	11.2	15.0	18.7	"	30	2.5	15	6.0	65	---
37, 84	13.5	18.0	22.5	"	30	"	14	8.0	55	---
38, 85	16.5	22.0	27.5	"	30	"	13	10.0	50	---
39, 86	20.2	27.0	31.3	"	30	"	11	11.0	45	---
40, 87	24.7	33.0	41.2	"	30	"	10	13.0	43	---
41, 88	29.2	39.0	48.8	"	30	"	9	14.0	42	---
42, 89	35.2	47.0	58.8	"	30	"	8	15.0	40	---
43, 90	42.0	56.0	70.0	"	30	"	7	19.0	36	---
44, 91	51.0	68.0	85.0	"	30	"	6	24.0	32	---
45, 92	62.0	82.0	102.0	25	33	"	5	31.0	28	5.6
46, 93	75.0	100.0	125.0	25	33	"	4	40.0	25	5.0
47, 94	90.0	120.0	150.0	25	33	"	3	50.0	22	4.4

1/ Test frequency range: 25 through 150 MHz.

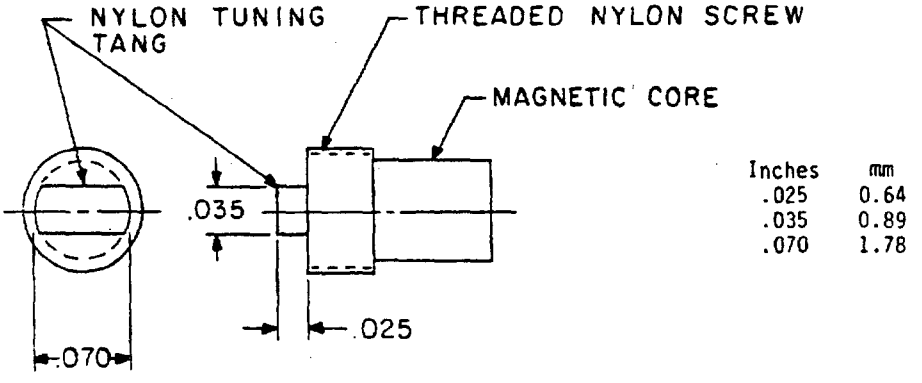
For electrical characteristics measurements use TF-250RX-3 test fixture, or equal (see figure 4). Fixture inductance (approximately 0.009 µH) should be subtracted from calculated inductance.

Test frequency range: 2.5 through 25 MHz.

For electrical characteristics measurements use TF-260Q-3 test fixture, or equal (see figure 5). Fixture inductance (approximately 0.028 µH) and residual Q-meter inductance (approximately 0.01 µH) should be subtracted from indicated inductance.

2/ Maximum current allowed not to exceed the specified temperature rise.

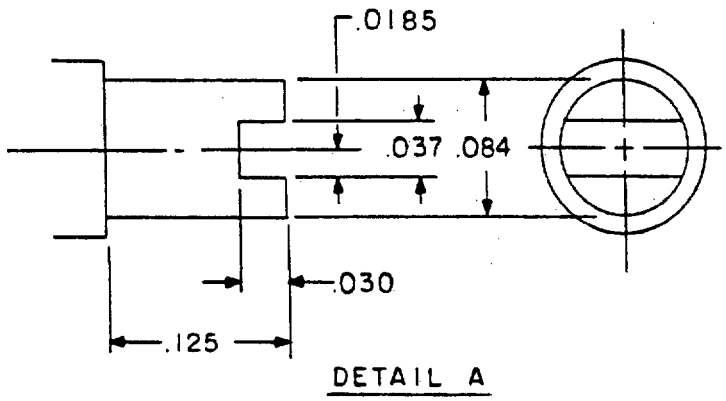
3/ Applicable to ferrite core only.

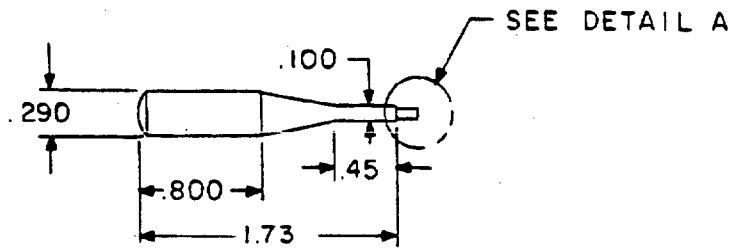


NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

© FIGURE 2. Core and adjustment screw assembly.

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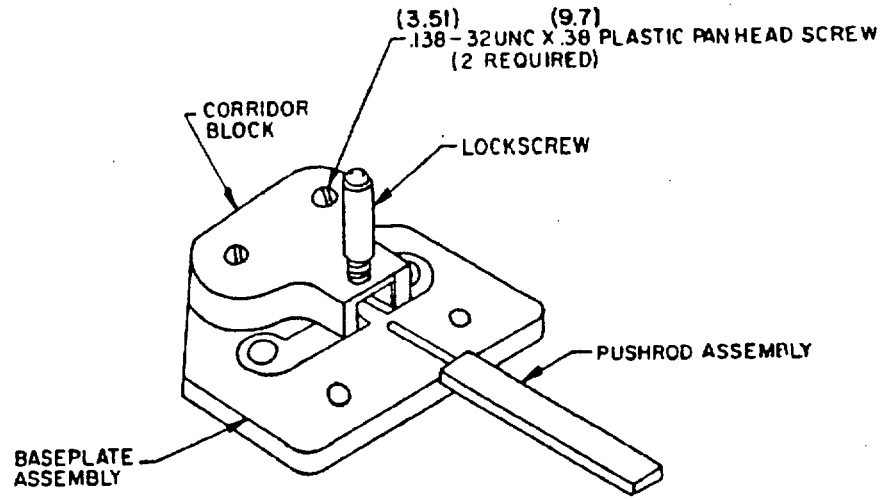




NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010.
4. Material: plastic.

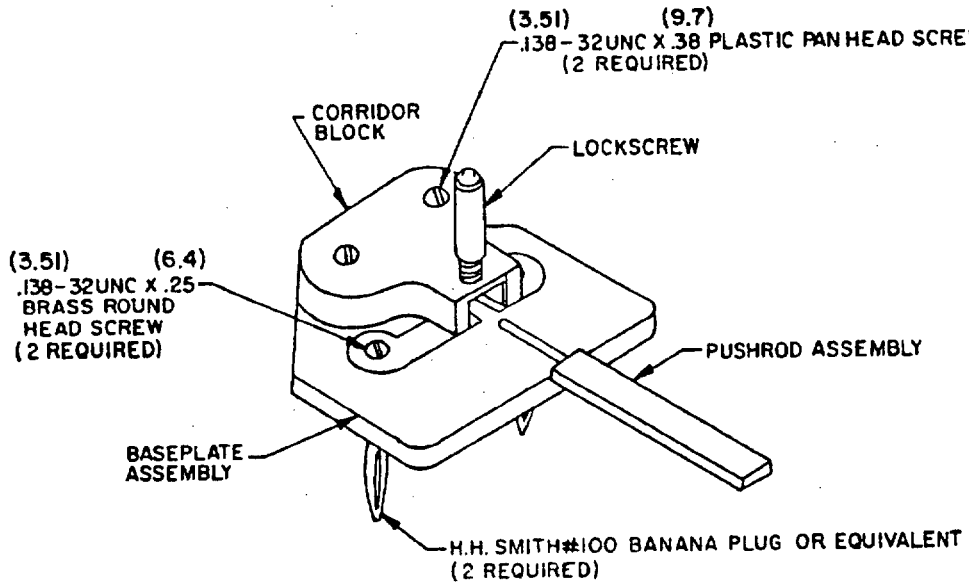
© FIGURE 3. Tuning tool (TT-2271 or equal).

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 4. Typical chip coil test fixture (TF-250RX-3, or equal).



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 5. Typical chip coil test fixture (TF-2600-3, or equal).

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TABLE II. Electrical characteristics (final).

Inspection group	Allowable variation from the initial measurements			
	Inductance	DC resistance	Self-resonant frequency	Q
Qualification inspection	Percent		Percent	Percent
Group II	±5	±(3% +0.001 ohm)	-8	-10
Group IV	±5	±(2% +0.001 ohm)	-10	-10
Group V	±2	---	---	-10
Quality conformance inspection				
Group C				
Subgroup II	±5	±(3% +0.001 ohm)	-8	-10
Subgroup IV	±5	±(3% +0.001 ohm)	-8	-10

Supersession data: A new part numbering system with codes for termination materials was incorporated with revision A, superseding MIL-I-83446/9, dated 1 March 1979.

M83446/09- (dash number from table I) C supersedes M83446/9- (dash number from table I).

Examples:

M83446/09-21C supersedes M83446/9-21.
M83446/09-03C supersedes M83446/9-3.

Part number: The part number shall be in the following form.

M83446/09- 04 B

- Prefix and specification sheet number
- Sequentially assigned dash numbers (see table I)
- Termination finish (see MIL-C-83446)

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - AS
Air Force - 19
DLA - ES

User activities:
Navy - OS
Air Force - 11

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5950-0685)

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